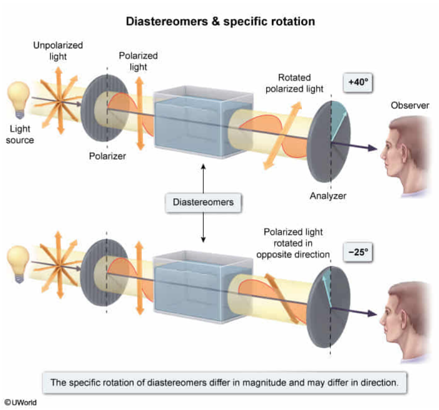
Which term can be used to classify the relationship between two isomers that have the same connectivity but specific rotations of +40° and −25°, respectively?

- A. Enantiomers [16%]
- B. Racemic mixture [9%]
- ✔ C. Diastereomers [44%]
- D. Conformational isomers [28%]

Correct
44% answered correctly

Explanation:



Specific rotation, the degree to which chiral molecules rotate plane-polarized light, is unique to each chiral molecule. The value of specific rotation consists of direction (+ or −) and magnitude (number of degrees). Clockwise rotations are designated as (+), whereas counterclockwise rotations are (−).

Isomers are molecules with the same molecular formula but have either different connectivity (constitutional isomers) or spatial orientation (stereoisomers). Diastereomers **differ** at one or more stereocenters but have the orientation of at least one stereocenter in common. The specific rotations of diastereomers differ in **magnitude** and may differ in **direction**. The same is true for constitutional isomers.

Because the question states that the two molecules have the same connectivity, they cannot be constitutional isomers. Because they have different specific rotations (direction and magnitude), it can be concluded that the two molecules are diastereomers.

(Choice A) Enantiomers have specific rotations of *equal* magnitude but *opposite* directions. If the molecules were enantiomers, they would either have specific rotations of +40° and −40° or +25° and −25°.

(Choice B) A racemic mixture is composed of 50% of one enantiomer and 50% of the other. Racemic mixtures have a specific rotation of 0° because the specific rotations of the two isomers cancel each other. Each isomer in the question has a nonzero specific rotation and therefore cannot be racemic.

(Choice D) Conformational isomers are versions of the same molecule that form when an atom rotates about its bond. Because conformational isomers are forms of the *same* compound, they will have the *same* specific rotation rather than different specific rotations.

Educational objective:
Specific rotation is the degree to which a chiral molecule rotates plane-polarized light. The value of specific rotation contains direction (+ is clockwise, − is counterclockwise) and magnitude. Diastereomers differ in the magnitude of their specific rotations (and may differ in direction), whereas enantiomers have specific rotations of the *same* magnitude but *opposite* direction.

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